



NANYANG
TECHNOLOGICAL
UNIVERSITY

ML in Algorithmic Trading/ NLP in Finance

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Outline

- How is NLP used in Finance
 - Sentiment analysis in finance
- What is Algorithmic Trading?
 - Technical Indicators in Finance
 - How TA is enhanced with AI/ML

Alternative Data in Finance

- Financial trading uses lots of data:
 - Market data
 - P/E, P/B ratios etc. (largely numerical in nature)
- New types of data –
 - Unstructured data (text)
 - Company reports & news
 - Social media
 - Imagery data (satellite data)
 - Sound data (voice transcripts)

Text with Natural Language Processing

- NLP is the computational quantification of text with AI to elicit insights like 'what', 'where', 'who' from unstructured data eg text
- *Precursor to LMMs*

TEXT ANALYTICS SERVICE

Intelligent Tagging

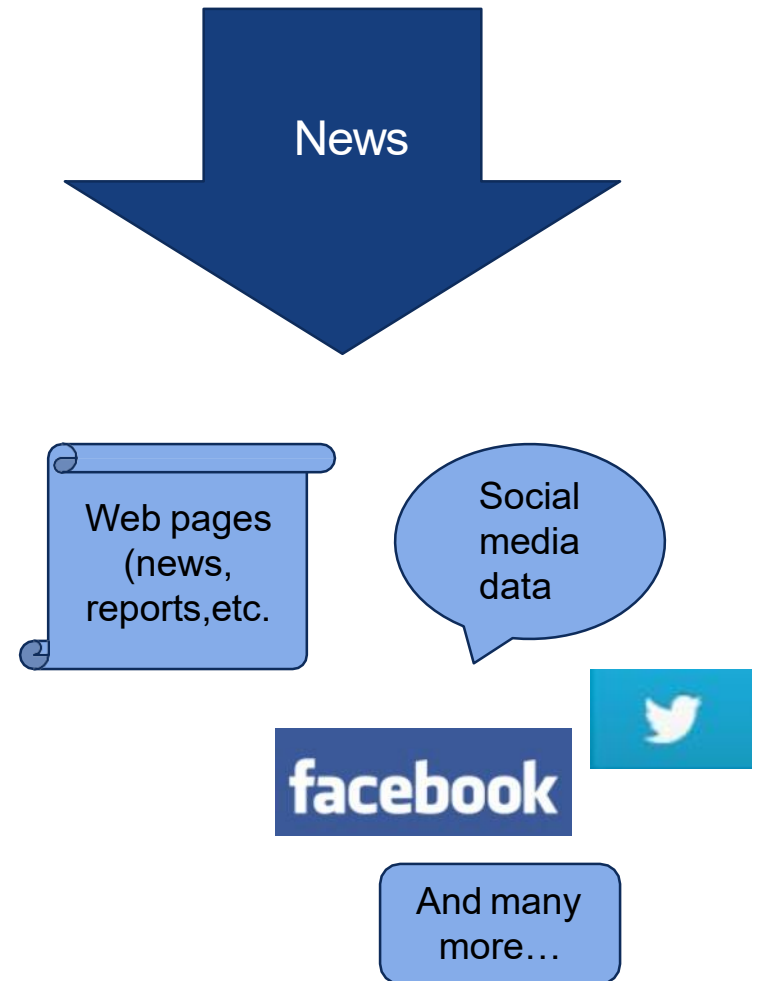
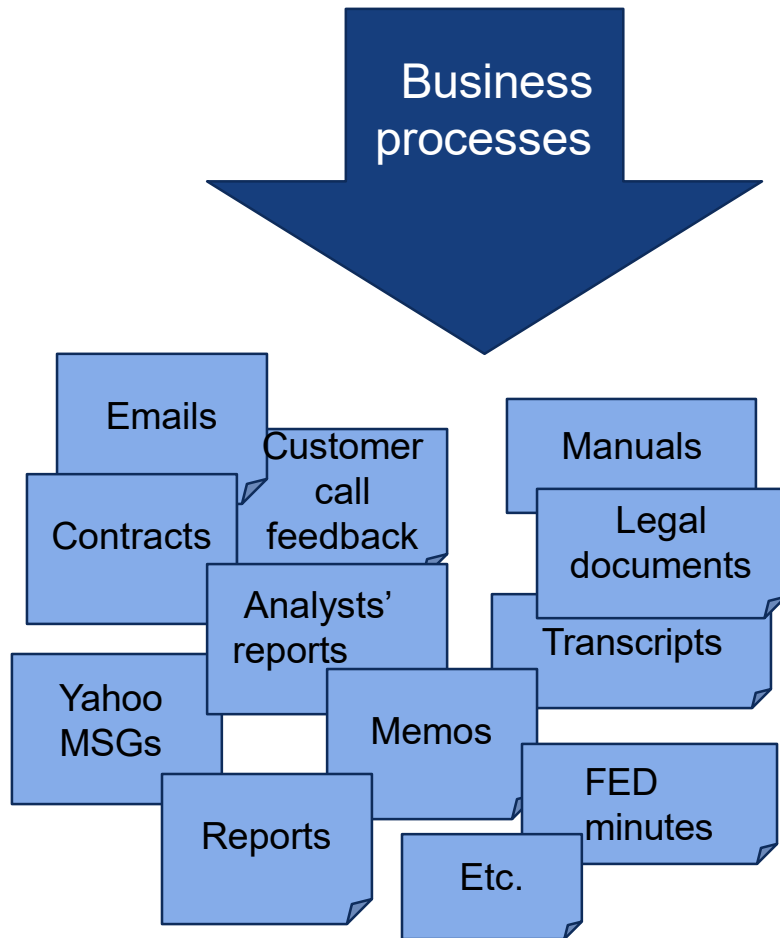
Using natural language processing, text analytics, and data-mining technologies, Intelligent Tagging turns large amounts of unstructured data into precise advantage.

[Read more on Refinitiv Intelligent Tagging >](#)



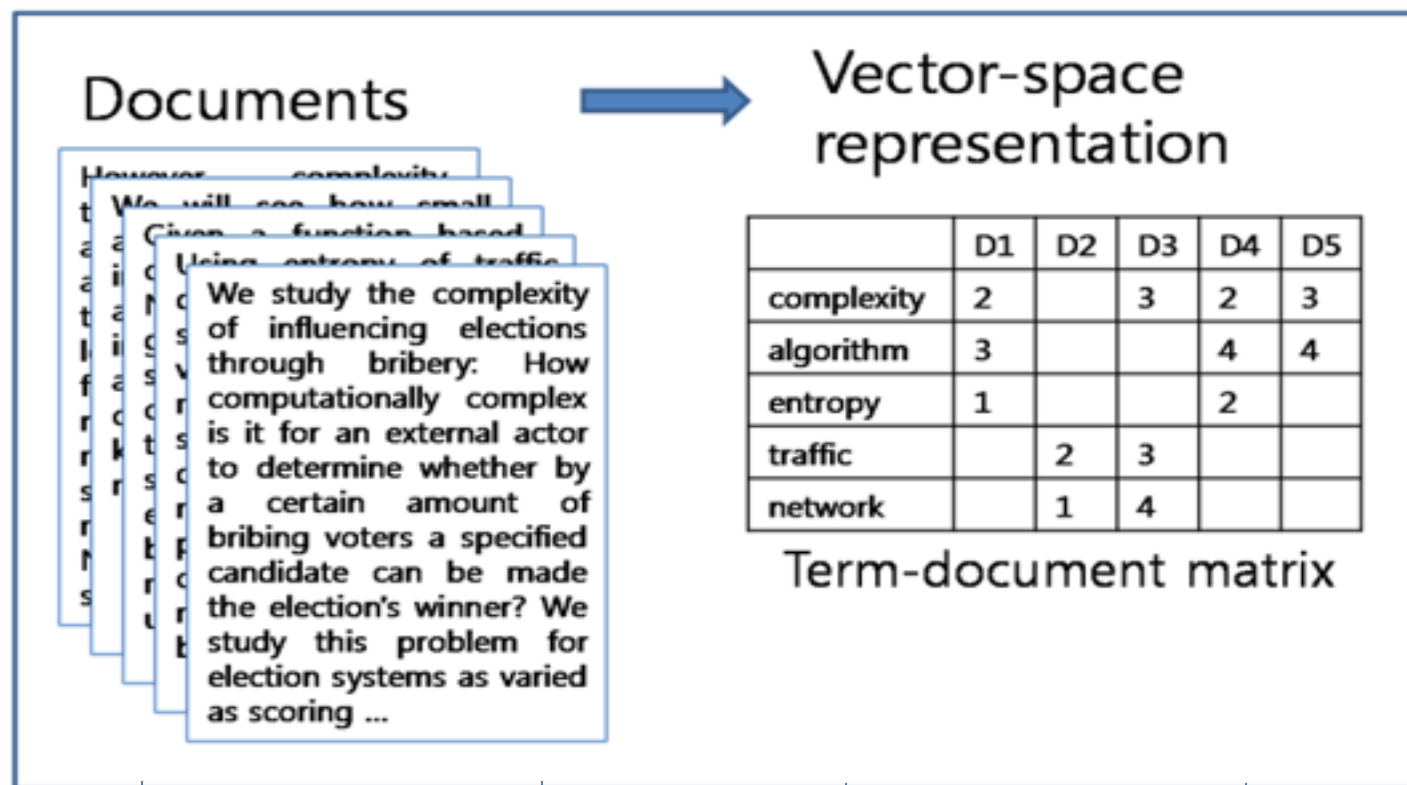
Intelligent Tagging overview (1:56)

Sources of Text Data



Text as Data

- Computers read only numbers. How do you convert the text into numbers?



Convert Text to Numbers :

Bag of Words Approach

- BOW: Bag of words. Simple counting of no of words in the documents to form a term-document matrix
- Pros:
 - Efficient: words surmise the meaning of the docs
 - Cons: language is only language if the words are used in a particular order. This order is lost

Tokenisation

- Use tokens not words. Process called tokenization. Breaks down large text into smaller manageable (numerical) units.
- Into sub-words, root words (lexeme), or character tokens.
 - Eg: running -> run – ning
 - Best, better, good -> good
- Enables AI to understand language/ be efficient across word -> how LLMs break down text into data

Word Embedding

- A word is recognized by its company. Also called word embedding.
- Key is how to represent a word with ‘numbers’/ vectors? Dependent on its context.



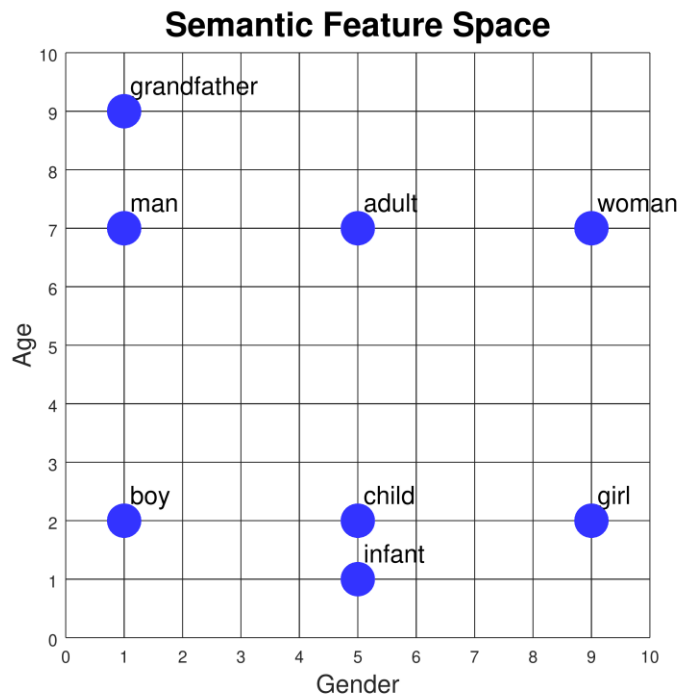
"You shall know a word by the company it keeps..."
– John R. Firth (1957)



"... the complete meaning of a word is always contextual, and no study of meaning apart from context can be taken seriously"
– John R. Firth (1935)

Word Vectors

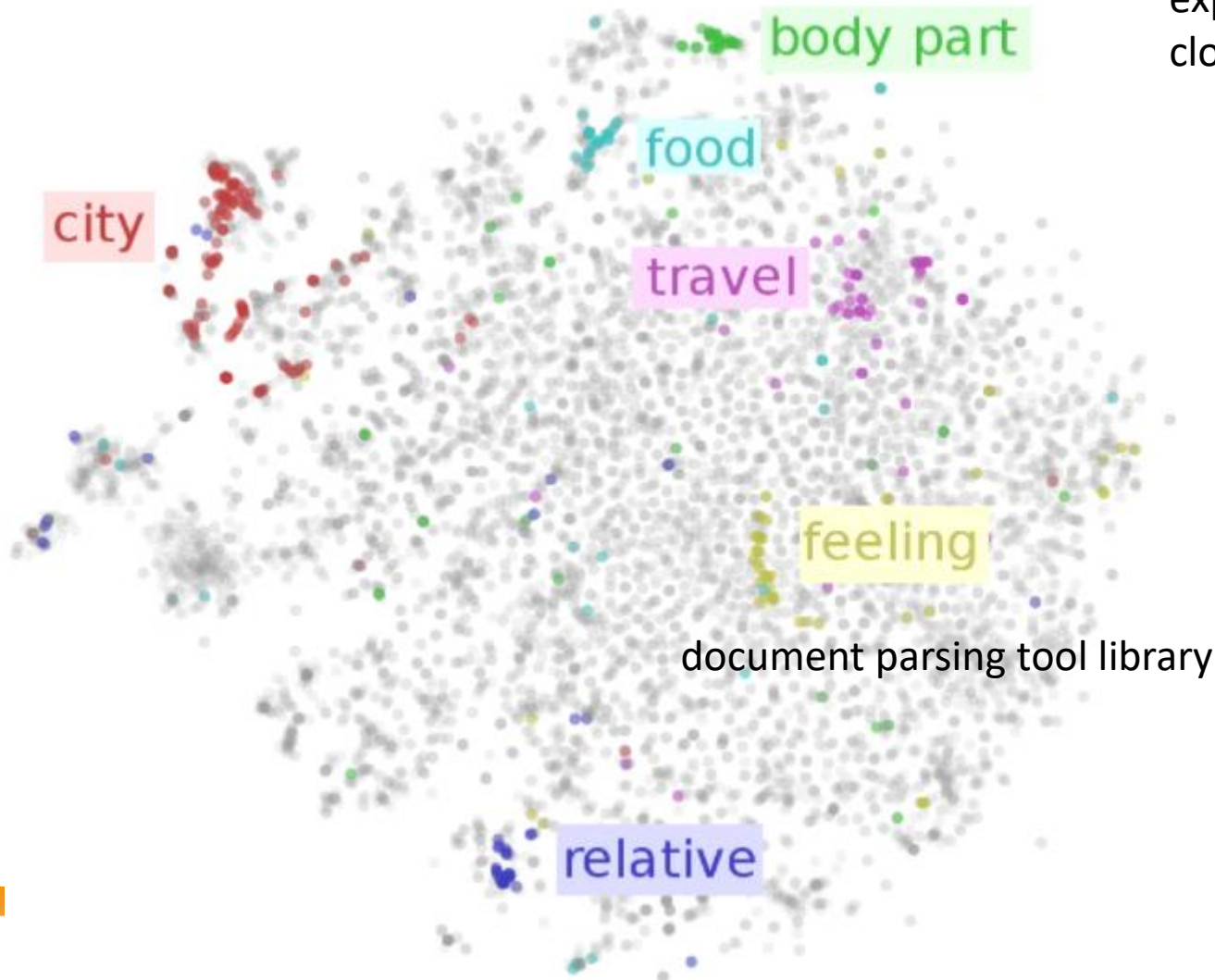
- How to represent tokens (words) into numbers that computers crunch?
- Becomes vectors of numbers.



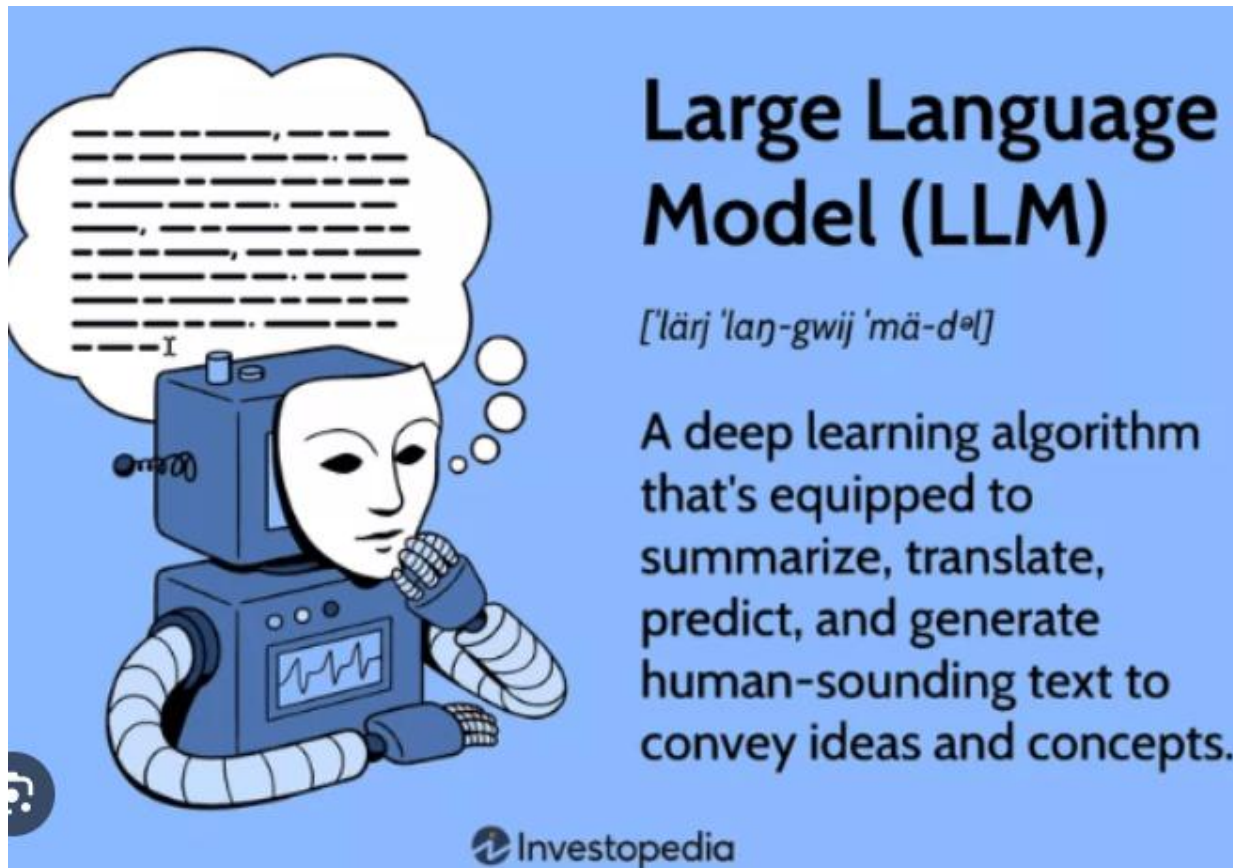
Word Coordinates		
	Gender	Age
grandfather	[1,	9]
man	[1,	7]
adult	[5,	7]
woman	[9,	7]
boy	[1,	2]
child	[5,	2]
girl	[9,	2]
infant	[5,	1]

Represent the word as vectors into N-dimensional space

For eg., you would expect body part to be closer to food in 'space'



Large Language Models (LLMs)



LLMs are
GenAI
techniques

[LLM explained briefly](https://www.youtube.com/watch?v=LPZh9BOjkQs)

[https://www.youtube.com/watch?v=LPZh9BOj](https://www.youtube.com/watch?v=LPZh9BOjkQs)

[kQs](https://www.youtube.com/watch?v=LPZh9BOjkQs)

Different types of LLMs (2025)

What are they good for? And what do you use?

LLMs

Choosing a model based on your use case.



Jafar Najafov

GPT-5

the best all-rounder

Great at writing

Strong at coding

Excellent reasoning

Claude 4

deep reasoning + long context

Thinks through nuance & ethics

Great for careful instruction-following

Huge context window (≈1m+ tokens)

Gemini 2.5

real-time knowledge + multimodal

Strong search integration

Excellent logic & math

Combines text + images

Grok 4

an AI with personality

Real-time context via X

Edgier, more entertaining tone

Good for current events & chatter

Mistral 7B / Mixtral

speed + control

Lightweight & fast

Open-source & self-hostable

Cheap, private, customizable

Llama 4

a free, powerful base model

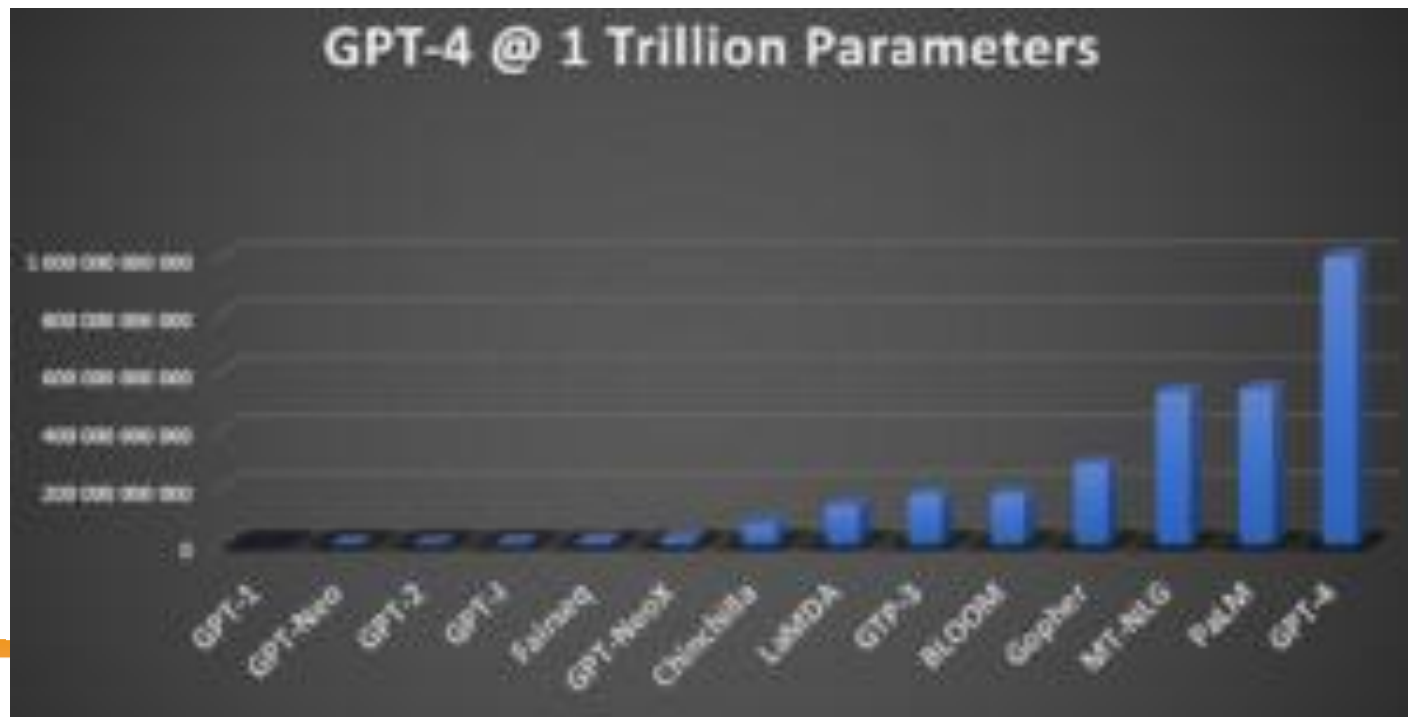
Open and widely adopted

Rapidly improving via community

Avoid API costs

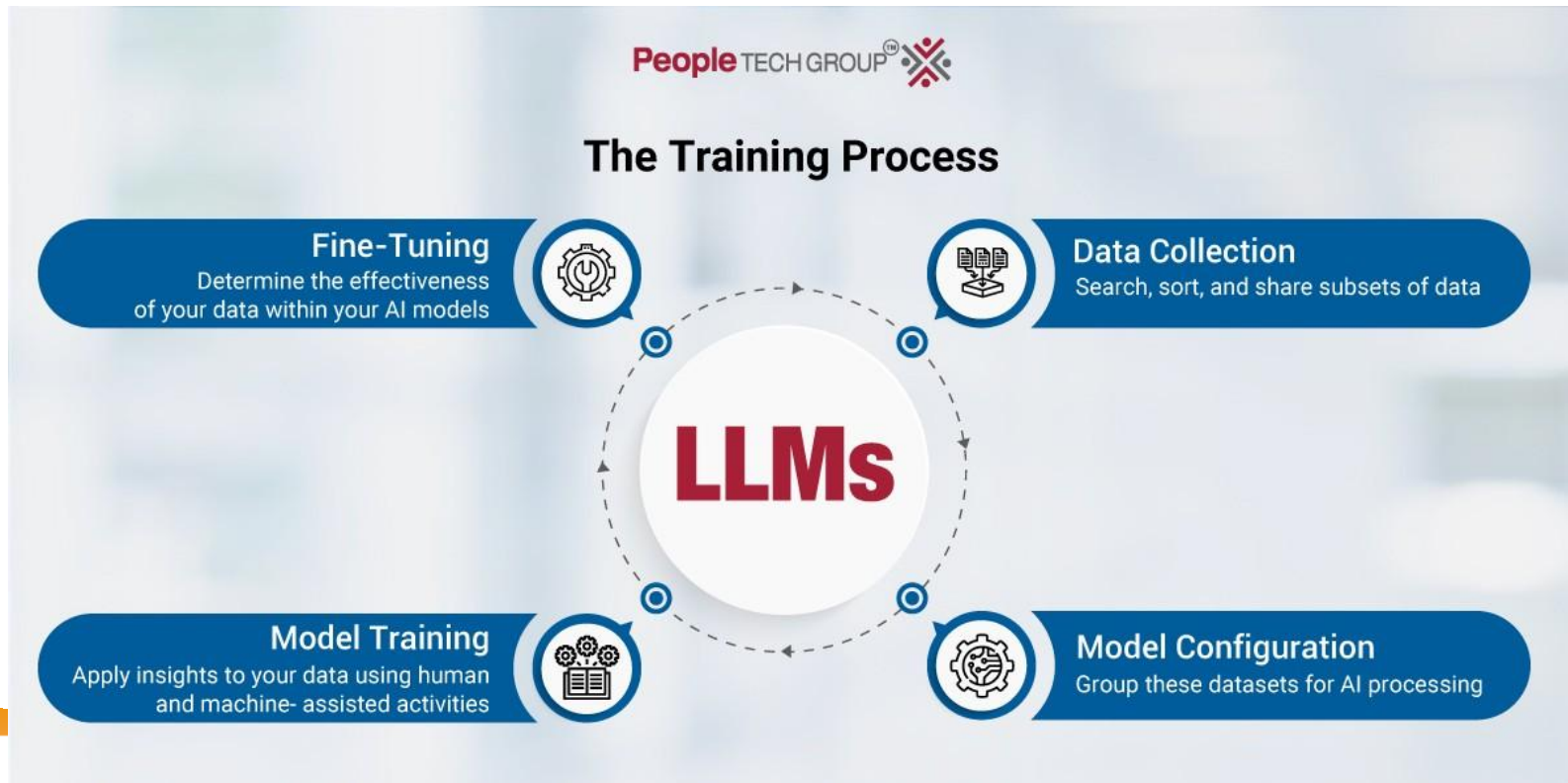
LLMs SIZE evolution (up to 2024(

- Very large deep learning models that are pre-trained on vast amounts of data.
- LLMs are a GenAI technique – generative AI with ever-growing number of parameters.



How do LLMs train?

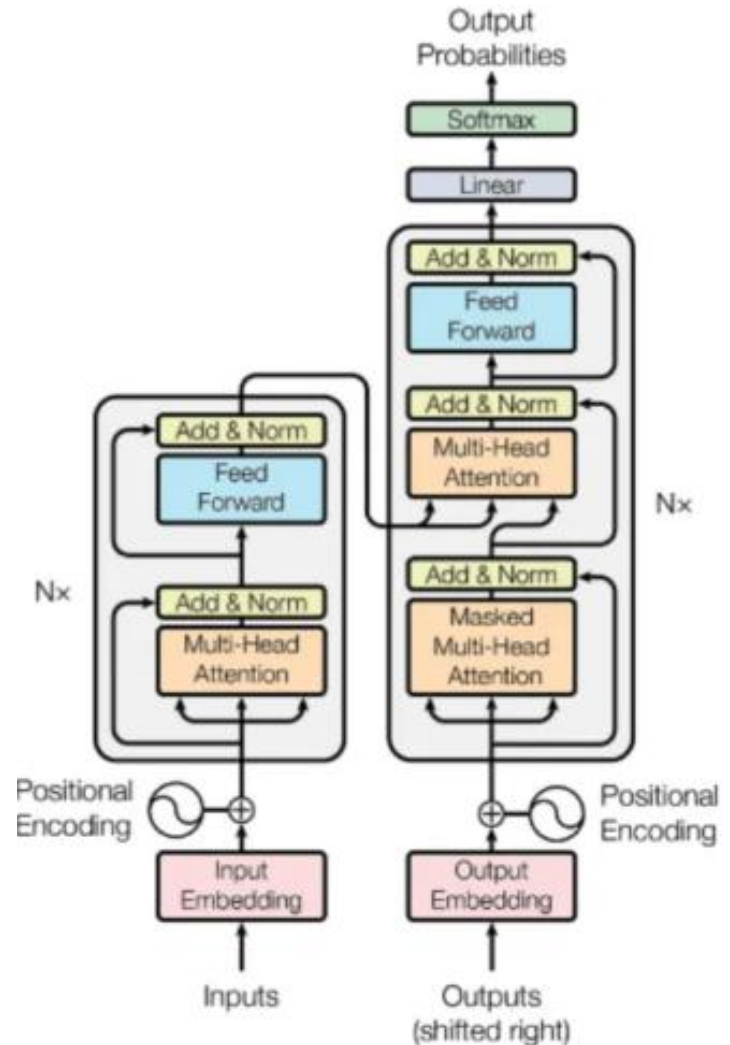
- Data collection – Internet
- Model configuration - transformers
- Model training – optimized weights (parameters)
- Fine-tuning – human feedback in the loop



Underlying technology - Transformers?

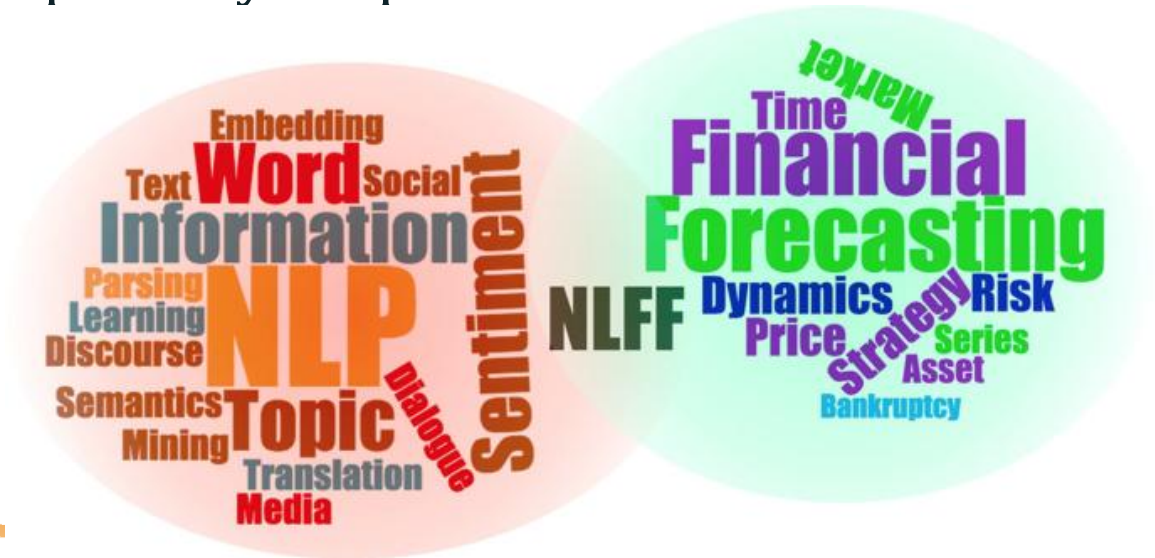
- A transformer is a type of artificial intelligence model that learns to understand and generate human-like text by analyzing patterns in large amounts of text data.

Key underlying
technology for LLMs.
Very complex!



NLP uses in Business Finance

- Huge use in asset management, compliance, trading, personal banking assistant (chatbots) and RPA.
- Many mundane tasks for eg in compliance. Names/ entity matching to blacklist, adverse media screening, credit analytics.
- Eg. How do traders possibly keep track of all the news incoming?



Sentiment Analysis (Market and Consumer)

- Markets move by behavioral sentiment – through news.
 - Fear & Greed
- Customer sentiment and feedback:
 - Eg. Grab

✦ AI Overview

Grab uses consumer sentiment to improve safety and customer experience through AI and social listening, and to drive business decisions like adapting to market trends and creating new features. It analyzes both positive and negative feedback from sources like social media, reviews, and surveys to understand user needs and concerns, which then informs improvements in existing services and the development of new features, such as the Super App model. [🔗](#)

For improving user experience and safety

- **Social listening:** Grab uses tools like Sprinklr to monitor social media for customer feedback, helping them to identify issues like reckless driving and misinformation. [🔗](#)
- **Proactive safety measures:** By listening to sentiment around safety concerns, Grab can implement and improve features like facial recognition for driver/passenger verification, the "Share My Ride" option, and trip monitoring to automatically detect potential issues, notes [AMBA-BGA](#). [🔗](#)

Asset Management

- Intelligent Tagging:
 - Making data smarter: using natural language processing, text analytics, and data-mining technologies, Intelligent Tagging turns large amounts of unstructured data into precise advantage.
- Get to know what events are affecting your portfolio, clients and acquisition targets etc.

<https://permid.org/tagging>

TEXT ANALYTICS SERVICE

Intelligent Tagging

Using natural language processing, text analytics, and data-mining technologies, Intelligent Tagging turns large amounts of unstructured data into precise advantage.

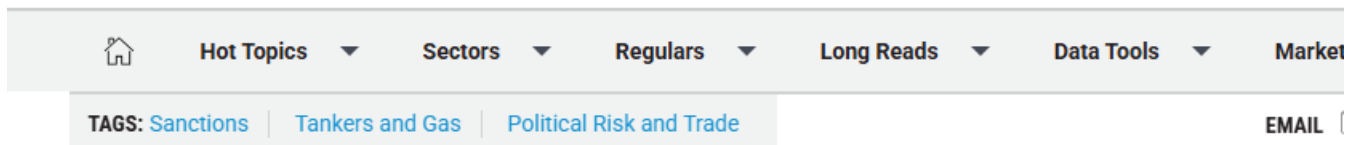
[Read more on Intelligent Tagging >](#)



Intelligent Tagging overview (1:56)

KYC/ AML

- Name checks in KYC/ AML: to verify the identity of a business or entity to prevent fraud, money laundering or in Politically exposed person. Or combat terrorism.
- Not trivial. Permindex.org for e.g. initiative by LSEG
- To escape international sanctions in trade financing:



Over 10% of the tanker fleet is now sanctioned following latest EU and UK measures

- *Latest EU sanctions package doubles the number of sanctioned ships with UK adding more, taking the total to 706 ships globally sanctioned*
- *Brussels and London forge ahead with Russia sanctions without US support*
- *Sanctions are now moving away from oil price cap compliance towards illicit and dangerous shipping practices as a rationale for designation*

Intelligent Tagging Demo

Answers questions like what, who, where and types of relationships in articles.

**LSEG** DATA & ANALYTICS

PermID

Tools APIs About PermID

PermID

Overview Entity Search Record Matching Intelligent Tagging Developer Community

Home > Intelligent Tagging > Results

FOUND IN DOCUMENT

^ ☒ ENTITIES ?

Relevance

☒ City

☒ IndustryTerm

☒ Region

^ SOCIAL TAGS

Relevance

2000s energy crisis100%

Price of oil100%

Pricing100%

West Texas Intermediate66%

Brent Crude66%

2017–2019 world oil market chronology66%

DOCUMENT VIEW

 Upload Again  View RDF

He added that there were substantial communications between the two sides over the weekend and more meetings were expected.

Brent crude futures rose 18 cents, or 0.28%, to \$63.50 a barrel by 0000 GMT, while U.S. West Texas Intermediate crude was at \$59.65 a barrel, up 16 cents, or 0.27%.

In the previous session, Brent settled 0.9% higher, and U.S. WTI closed up 1%.

The prospect of improved trade relations between the world's two largest economies has historically buoyed oil markets, as investors anticipate stronger global growth and increased fuel demand.

However, recent developments, including Beijing's expanded export controls on rare earths and Trump's threats of 100% tariffs and software export restrictions starting November 1, have weighed on sentiment. Last week, oil prices posted weekly losses and touched their lowest levels since May.

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Refinitiv Framework: Not trivial!

- Permid <https://permid.org/>
- Not trivial. Suppose KYC on Asia Pacific Brewery on customer database suspected on Blacklist.
- But who is Asia Pacific Brewery?
- Search results indicate multiple <https://permid.org/entity-search;search=Asia%20pacific%20brewery>
- Data entries in different offices regionally may have no standard way to data entry.

Topic Classification

- Supervised learning method to classify text into its relevant topics. The idea is that common topics tend to have common words. For eg. In finance, lingo like interest rates, trading, options, derivatives are common.
- The machine learns these words from a reference text, and then apply to new texts to classify them. In finance, category of news, IPO, compliance, bond raising etc.



Topics Relevance

- Knowing the news topics of companies allows you to keep abreast of news types of your competitors, investments, customers etc. and take according action.

BT The Business Times

[Singtel unit prices S\\$160 million in fixed-rate notes at 2.726%](#)

[SINGAPORE] Singtel Optus, a wholly owned subsidiary of the telecommunications giant, on Monday (Jul 21) priced S\$160 million in fixed-rate...

21 Jul 2025



NTU Nanyang Technological University - NTU Singapore

[Singaporean commodity trader taps Mauritius to finance Africa operations](#)

Singapore-based agribusiness group Valency International has raised US\$15m through a bond sale in Mauritius to finance its operations in...

29 Jul 2025



Which companies have recently issued new bonds?

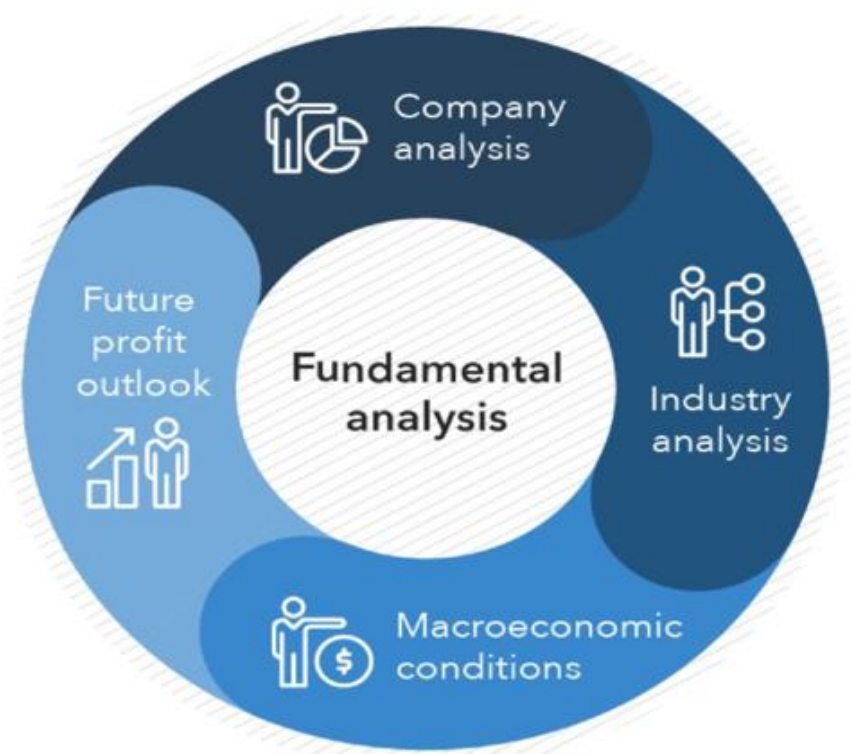
Personal Finance

- Relationship managers need to obtain sources of information –
 - investment products,
 - risk/ returns level etc. to match risk preferences
 - Client Q&A and chatbots
 - Build portfolios

What are the advantages and disadvantages of using LLMs for the relationship/ wealth managers?

Comparing between Fundamental and Technical Analysis

- Financial markets analysis uses either fundamental or technical analyses to predict market movements.
- AI/ML can be used with both fundamental and technical indicators for prediction, instead of human intuition.



What is technical analysis?

- Technical analysis is a method of predicting the future direction of a market's price by studying historical chart patterns and formations.
- The aim of technical analysis is to identify *recognisable patterns* to help traders find the right time and price point at which to enter and exit the market.

**Formulated by Charles Dow,
1st editor of WSJ in late 1890s
and based on Dow theory**

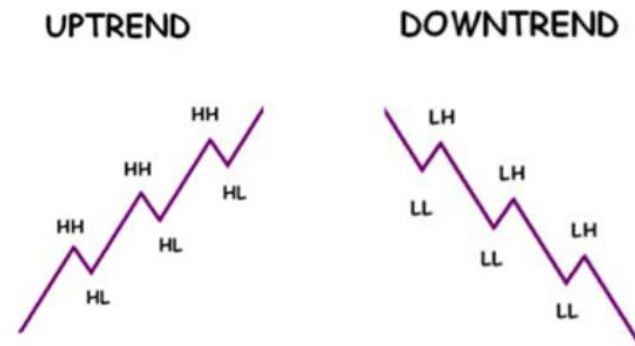


History does not
repeat but it rhymes

Based on Dow Theory

Hypothesis in theory:

1. The averages discount everything (from demand and supply)
 - Before that *individual prices*...
2. The primary trend is inviolate
 - Trends consist of tide (primary), wave (secondary) and ripples (minor)
 - Uptrend/ downtrend:



Trends in Dow Theory

Other trends: secondary trend

- Corrective phase of the primary trend (retrace $\sim 1/3$ to $2/3$ of primary trend in 3-4 months)
- Minor trends: ($\sim < 3$ weeks) fluctuations in secondary trend.

Primary trend

- Accumulation phase
- Public participation phase
- Distribution phase

Primary Trends in Dow Theory



Accumulation Phase: (consolidation)

- Occurs after a substantial decline in prices.
- Typically marked by negative sentiment, bearish media coverage, and frantic selling by uninformed participants.
- Informed investors take advantage of low prices to accumulate shares gradually, anticipating a bullish reversal.

Trend Phase:

- Triggered by expectations of higher prices post-accumulation or lower prices post-distribution.
- Begins with technical breakouts signaling an uptrend, attracting early adopters and savvy investors.
- Media coverage becomes optimistic, drawing increased public participation.
- Participants exhibit regret bias, buying on dips and covering short positions as prices rise.

Distribution Phase:

- Follows a significant price increase, accompanied by positive company data and bullish media coverage.
- Uninformed participants exhibit bullish sentiment, leading to a buying frenzy and rising margin debt.
- Informed investors initiate selling campaigns.

Technical Analysis is widely used

Monday, August 22, 2022

THE BUSINESS TIMES



CHARTPOINT

Crude palm oil heading towards downtrend consolidation phase

By Avtar Sandu

CRUDE palm oil futures rose sharply over the last 2 weeks from lows of RM3,700 (\$51,150) a tonne, propelled by gains in soybean oil on the Chicago Board Of Trade (CBOT) and expectations of better exports for the month of August. The world's largest producer and exporter, Indonesia, recently announced an increase in export taxes for shipments for the week of Aug 9 to Aug 15, giving Malaysian crude palm oil (CPO) priced in ringgit a competitive edge.

Owing to ideal rainfall and favourable weather conditions, a surplus in palm oil production is expected to hit markets in August and September as the seasonal peak production season gets underway. Inventories of top exporter

Indonesia have swollen to 10 million tonnes compared to the average of 5 million tonnes, fully utilising capacity storage in tanks, wells and barges. After a drop from highs by 45 per cent, Indonesian palm oil is currently the cheapest vegetable oil in market.

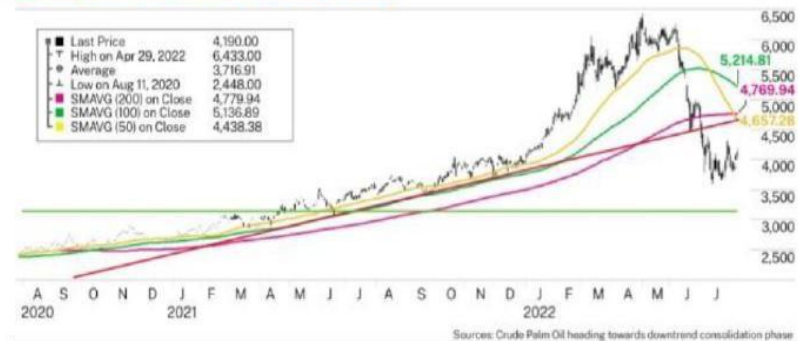
Looking at the weekly chart, palm oil double topped at levels of RM7,200 on Mar 9, 2022, followed by highs on Apr 29, just after the Indonesian government announced an export ban on Apr 28. The catastrophic move to safeguard local prices and counter the threat posed by ongoing geopolitical crisis that was expected to halt the global supply of soybean oil, sunflower oil and other vegetable oils, backfired by destroying local demand and increasing Indonesian stockpiles.

Soaring palm oil prices amid a potential recession along with the economic downturn in China threatened demand even further, resulting in a free fall in palm oil prices to the lows of RM3,700 in mid-July.

The historic high and low levels of RM3,000 to RM3,200, before palm oil started an uptrend in October 2020, continue to act as strong support. Market talk projects that the world's most consumed cooking oil may continue its slide towards RM3,000 to RM3,200 a tonne by September this year, driven by surging supplies from Indonesia.

The upward trend line joining lows since May 2020 has been breached and will now act as resistance whenever prices try to break through. In its recent rise, palm oil

Crude palm oil futures weekly chart



touched the 50 day moving average at 4,438 and for the next few weeks it should consolidate between a larger range of 3,700 to 5,000 without the intervention of any market-moving events.

But are technicals sufficient to guide volatile movements in agricultural commodities? Demand-supply dynamics have a part to

play as well. The world's production of 7 oilseeds is estimated at 610 million tonnes, up by 33.4 million tonnes for 2023, with supplies expected to pour in during the first quarter of 2023. Ongoing geopolitical crisis will pose constant bullish threat as supply chains look restricted; we should witness some recovery in prices before soybean

oil, corn oil and rapeseed oil supplies from the southern hemisphere hit the market in 2023. There is also the possibility of the resumption of the flow of sunflower oil from the Black Sea region in the likelihood of a ceasefire.

The writer is senior commodities manager, Phillip Nova

Why does it work (or not)

Pros

- Relatively 'easy' to pick up
- Range of tools for any security for (usu.) short term
- History rhymes: greed and fear (primitive emotions). Does it?

Cons:

- **'Voodoo' science. Not theory based?**
- Subjective: everyone thinks they are right. But even experts get it wrong!
- Limited information set from historical mkt data (ignore fundamentals?)

Two Main Types of Technical Analysis

- Key types of technical analysis:
 - Trend:
 - Price/ volume momentum indicators
 - Fibonacci sequence
 - Overbought or oversold indicators
 - Support/ resistance levels
 - Moving averages convergence divergence

Augmented with AI/ML + fundamental analysis ->
part of arsenal for informed decision

Moving Averages

Moving average (MA) Tools:

-Technical indicator in finance that smooths out price fluctuations by calculating the average closing price over a specific period.

- Simple MA
- Exponential MA

Average out over last 50, 22 days etc. (user decide)

For e.g. you can use ML to determine the best parameters according to backtesting.

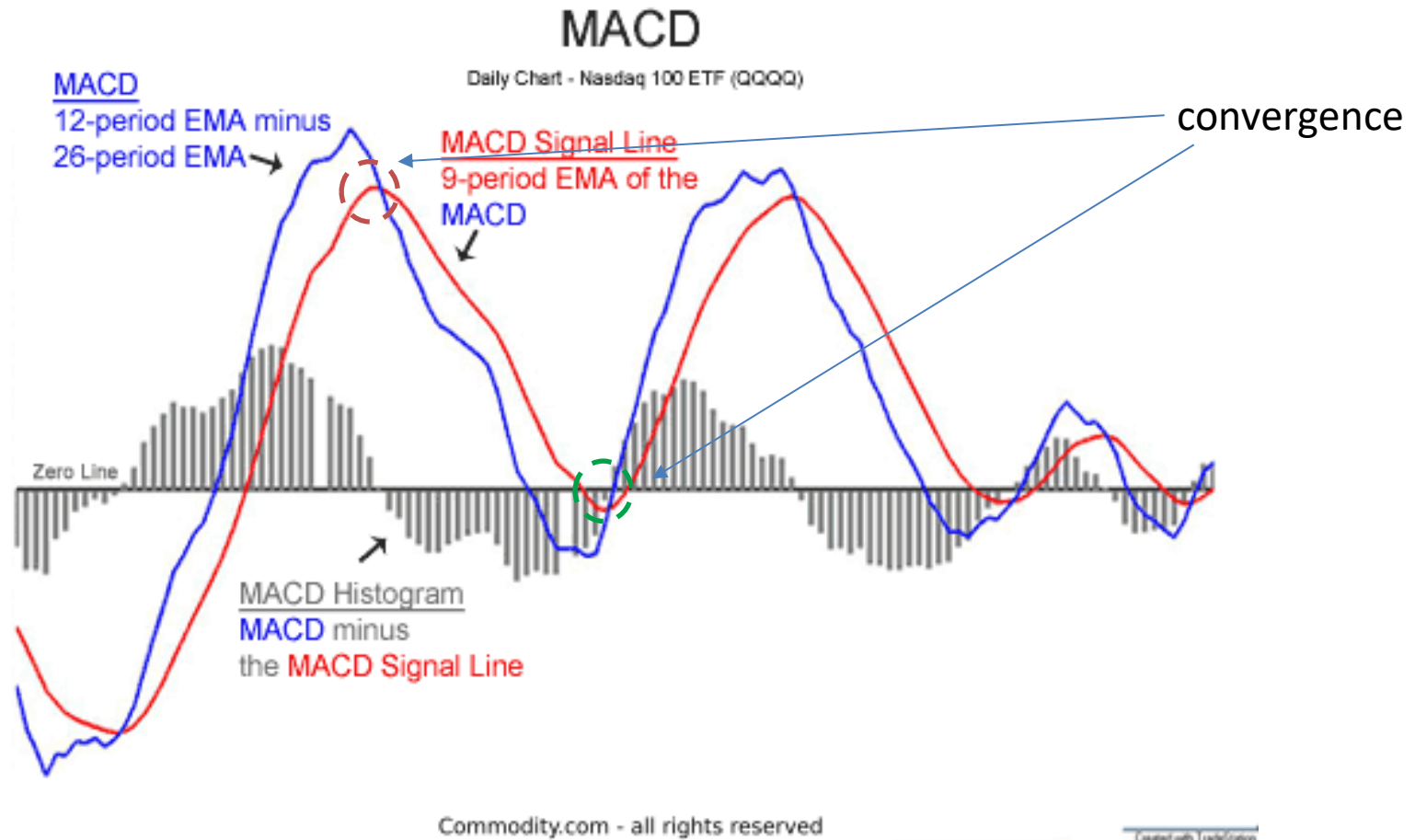
Uses:

- Indicates trends/ support levels
- filter out market noise

Moving Average Convergence Divergence (MACD)

- MACD is a technical analysis indicator used to identify the momentum and direction of a trend.
- Comprises 2 technical indicators
 - MACD line = 12 period EWMA - 26-period EMA
 - Signal line = 9-period EMA of this MACD line
- Stages:
 - *Convergence (crossover)*: **MACD** line **crosses above** the **signal line**, it's considered a *buy signal*. When it **crosses below**, it's a *sell signal*
 - *Divergence*: When the **MACD line** diverges from the price, it can signal a potential reversal in the market.
- There is no reason to conform to the parameters set.

Moving Average Convergence Divergence (MACD)



Moving Average Convergence Divergence (MACD)



Bullish divergence

MACD forms 2 rising lows
while price forms 2 falling
lows -> **bullish**



Bearish divergence

MACD forms 2 falling lows
while price forms 2 rising
lows -> **bearish**

AI can be used to recognise these images in CV (computer vision)

Indicators & Oscillators: Relative strength index (RSI)

- Momentum indicator to measure speed and change of price movements.
- To determine overbought or oversold condition. Can be used to assess whether momentum will continue or a reversal would come.
- Between 0 to 100, formula:

$$RSI = 100 - \frac{100}{1 + \frac{n_{up}}{n_{down}}}$$

n_{up} - average of price gains during the n past days

n_{down} - average of price falls during the n past days

- Again, ML can be used to optimise the parameters in prediction.

Indicators & Oscillators: Relative strength index (RSI)



Credit: Tradingview

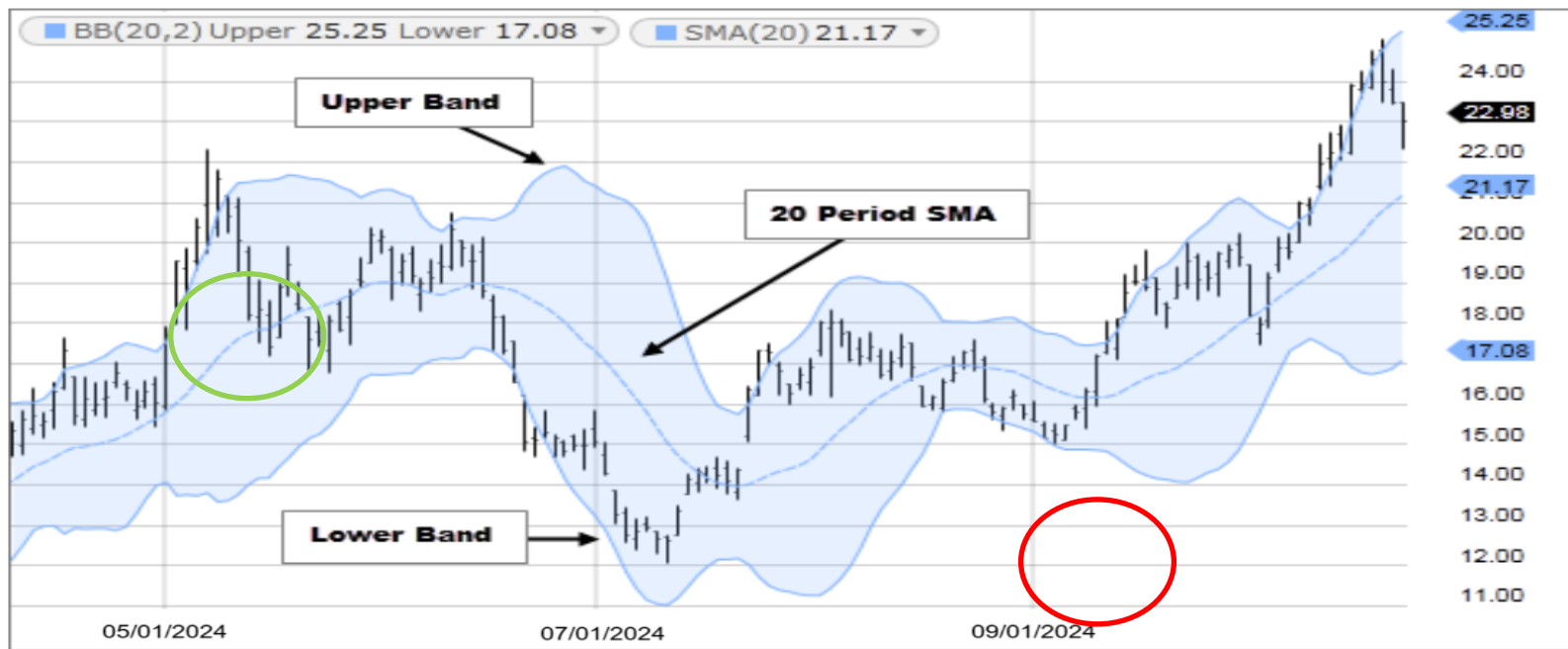


Like a gas indicator:

Typically
< 30 (oversold)
> 80 (overbought)

Indicators & Oscillators: Bollinger bands

- Determine oversold or overbought conditions. Indication of volatility.
- Consists: Middle line SMA + upper/ lower bounds (+/- 2 standard deviation)



Indicators & Oscillators: Bollinger bands

- Some 'words of wisdom'
 - If the band tightens, possibility of sharp up or down movements (trend)
 - If the band widens, existing trend could lead to consolidation. (ending trend)
 - If prices exceeds upper band -> sign of overbought
 - If prices breaks lower band -> sign of oversold
- These 'wisdom rules' can be written as part of predictors for ML algorithms.

Some practical advice

- Your trading horizon matters! Intraday? A few days? Usually technical trading not applicable for long term.
- Technical trading cannot predict events especially on fundamentals. It is largely based on behavior.
- Sometimes you don't have to buy/sell. Not trading is also an decision!

Many GenAIs projects are failing

LLMs form the backbone of GenAIs technology, but ...

<https://fortune.com/2025/08/18/mit-report-95-percent-generative-ai-pilots-at-companies-failing-cfo/>

NEWSLETTERS · CFO DAILY

MIT report: 95% of generative AI pilots at companies are failing



BY SHERYL ESTRADA

SENIOR WRITER AND AUTHOR OF CFO DAILY

August 18, 2025 at 6:54 AM EDT

Exercise: Discussion on Lessons Learnt

- Read and hear the article on Fortune.com link. What are the lessons learnt to enable successful deployment of GenAI projects in enterprises?
- How should (and can) that apply to your own experiences or company?



Case Study: Staying updated and Automating News on entities

- News:
 - Affect pricing in markets
 - Are forward-looking sentiment
- Corporate valuation:
 - Keeping track of news that affects fundamentals (cash flow channel – how much can firms earn?)
 - News moves discount rates (risk/discount-factor channel)

News queries help capture the dynamic inputs — both **hard fundamentals** and **soft sentiment signals** — that drive business valuation.



Case Study: Analysing Annual Reports

- Using OpenAI (ChatGPT) to analyse annual reports.
- This can be *automated* to *generate key insights* and for multiple reports.
 - Costs about 40 cents and 10 mins for one annual report on ChatGPT.
- Notebook location:
 - Why automate?
 - Increased productivity
 - Part of AgenticAI workflow



Conclusion

- **From NLP to LLMs:**

Natural Language Processing (NLP) converts unstructured text into numerical representations (tokens, word vectors, embeddings) that allow machines to understand language contextually, paving the way for modern Large Language Models (LLMs).

- **Customization and Future Trends:**

Enterprise-grade solutions use **RAG** and **MCP** to integrate LLMs with proprietary datasets and tools, addressing the limitations of generic models — though many GenAI pilots still struggle due to poor integration and governance.



Conclusion

- Technical analysis has its place as part of trading tools.
- It is relatively easy to use – but highly subjective and cannot be used to predict unexpected events. Use it with other tools in the arsenal – fundamentals, AI/ML.
- There is a wide array of tool within technical analysis
- Technical analysis could be based on behavioral factors in the finance theory.

Resources for Technical Trading

"Technical Analysis of the Financial Markets" by John J. Murphy

"Japanese Candlestick Charting Techniques" by Steve Nison "Technical

Analysis Using Multiple Timeframes" by Brian Shannon

"Technical Analysis for Dummies" by Barbara Rockefeller and Vicki Schmelzer

Quiz Time



Question: Understanding Data in Modern Finance
Which of the following best describes the evolution of data used in finance?

- A. Finance only relies on numerical data like prices and ratios
 - B. Financial analysis now incorporates unstructured data such as text, images, and audio
 - C. Only structured data is relevant because computers cannot process unstructured data
 - D. AI models ignore alternative data because it lacks numerical representation
- 

Question: Role of NLP and LLMs

What is the *main role* of Natural Language Processing (NLP) in finance before the emergence of LLMs?

- A. Generating fake financial reports
 - B. Converting textual data into numerical form that AI can understand
 - C. Calculating bond yields and portfolio returns
 - D. Encrypting trading algorithms
- 

Question: RAG vs MCP in Customizing LLMs What is the key difference between Retrieval-Augmented Generation (RAG) and Model Context Protocol (MCP)?

- A. RAG is a training algorithm, MCP is a visualization tool
 - B. RAG connects to external databases, MCP standardizes how LLMs access tools and data
 - C. Both are identical and interchangeable in GenAI systems
 - D. MCP is for small models, RAG is for large models only
- 